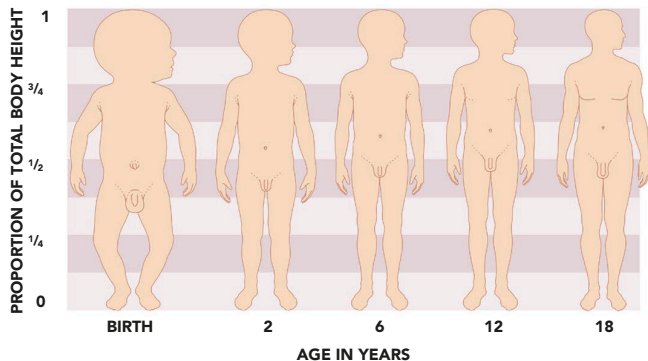


CHANGING PROPORTIONS

A newborn's head is relatively large, being wider than the shoulders and representing about a quarter of the baby's total height; the legs are about three-eighths of this height. As the child grows, the arms and

legs "catch up". At two years, the head is about a sixth of the total height. When final adult size is reached during adolescence, the head is only about an eighth of the body length, and the legs one-half.



HEAD-BODY PROPORTIONS

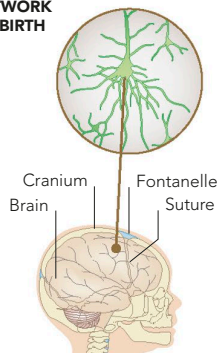
If the body's height at different ages is superimposed onto a grid, the changes in head-body proportions that take place from birth to adulthood are clearly shown. The overall growth trend is for the head to lead, growing first and fastest. Then the other regions of the body catch up: first the torso, followed by the arms, and finally the legs.

SKULL AND BRAIN

At birth, the brain is a quarter of its adult size. It has almost its full complement of neurons, but they have yet not made many interconnections. Gaps (fontanelles) and

seams (sutures) between the skull bones allow for expansion. By two years, the brain is four-fifths of its adult size, and neurons are forging links into networks.

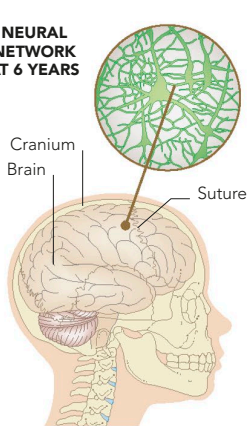
NEURAL NETWORK AT BIRTH



BIRTH

The cranium and brain are huge compared to the small facial bones. The neurons make limited links.

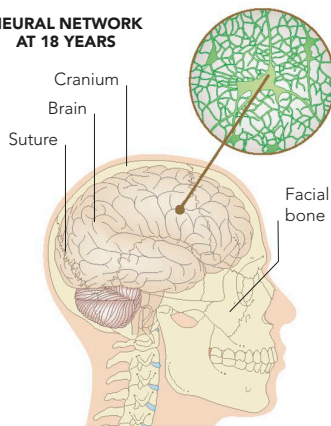
NEURAL NETWORK AT 6 YEARS



SIX YEARS

Cranial bones are fusing at the sutures. Neurons rapidly extend their projections and their links.

NEURAL NETWORK AT 18 YEARS



ADULT

The cranium is solidly fused, the brain is full-sized, and new neural links are made less often.